

Part 4 — Portfolio Construction and Risk

Part 4 turns the signals into long/short portfolios that could be traded, controls their risk, and applies a decision rule to the result. Every design choice was written down and committed before any result was computed (docs/part4_prereg.md). An independent code audit then found several defects; the fixes and their effects are listed in docs/part4_deviations.md. None of them changes the conclusion.

Reconciliation with Parts 1 and 3

Part 4 first reproduces the committed Part 1 and Part 3 numbers with the team's own functions, and does so exactly. The first version of Part 3 scored the signal against the return two sessions after the leader's move instead of one, and Part 1 already shows the lead-lag effect is gone after one session (Fama-MacBeth $t = 0.72$ at lag 2). Part 3 now uses the same one-day gap as Parts 1 and 2, and both versions reproduce:

Signal	Part 3, first version (two-day gap)	Part 3 as run now (one-day gap)
leader_ret_lag	0.56 bp/day (t 0.67)	3.36 bp/day (t 3.91)
conditional_signal	-0.12 bp/day (t -0.16)	2.98 bp/day (t 3.72)

results/p4_reconciliation.csv · Part 3 quantile_portfolios, top minus bottom quintile

At a one-day lag the conditional signal does sort returns in a naive quintile spread. The rest of this section asks whether that survives a properly built portfolio.

How the books are built

The leader signals (common_lag, shock_lag, leader_ret_lag and conditional_signal) take one value per industry per day, shared by every follower in that industry. An industry-neutral book on such a signal is identically zero, and the real breadth is the 48 industries rather than the ~1,750 followers. These signals therefore get an industry book. Each day the industries are ranked by the signal, and the ranks are made neutral to a constant (so the book is dollar-neutral), to the industry's average follower beta, and to the industry's average follower return on the previous day. Each industry's weight is then split equally across its followers. The last constraint mirrors Part 2's regression, which controls for the follower's own lagged return, and separates what the leader adds from the followers' own industry move. Betas are Dimson betas (same-day and lagged market, 252 sessions ending the day before, shrunk halfway to one).

The reversal benchmark (the negative of the follower's own lagged return) varies within industries, so it gets a stock-level book that is dollar-, industry- and beta-neutral. All books have gross exposure 1 (long \$0.50, short \$0.50) and are rebalanced daily. The neutrality constraints hold to floating-point precision.

Each book is evaluated under three timings. The headline is open-to-close: the book is formed from information at the close of $t-1$, bought at the open of t and sold at that day's close, which can actually be traded. Close-to-close, the timing of Parts 1 and 2, is an upper bound, because it requires trading at the same close used to compute the signal. Close of t to close of $t+1$ is a full session late, the timing of the first version of Part 3, and is used only as a check.

The conditional signal loses money once risk is controlled

Book	Timing	Return / yr	Vol / yr	Sharpe (SE)	t (NW)	Max DD
conditional_signal	open to close	-0.46%	2.3%	-0.20 (0.19)	-1.08	-16.6%
	close to close	-0.70%	2.5%	-0.28 (0.19)	-1.48	-21.7%
common_lag	open to close	0.85%	2.4%	0.35 (0.19)	1.90	-7.6%
	close to close	1.19%	2.7%	0.45 (0.19)	2.39	-8.2%
leader_ret_lag	open to close	0.79%	2.3%	0.34 (0.19)	1.80	-8.1%
	close to close	1.49%	2.6%	0.57 (0.19)	3.05	-10.3%
shock_lag	open to close	0.77%	2.3%	0.34 (0.19)	1.78	-6.5%
	close to close	1.32%	2.5%	0.52 (0.19)	2.74	-7.7%
reversal (benchmark)	open to close	2.52%	3.2%	0.78 (0.19)	3.70	-45.7%
	close to close	10.10%	3.7%	2.72 (0.19)	12.99	-16.4%

results/p4_performance.csv · gross-1 books, 1996–2024, Newey-West t-statistics with 5 lags

The conditional signal earns -0.46% a year open-to-close ($t = -1.08$) and -0.70% close-to-close ($t = -1.48$). None of the four pre-registered primary tests is significant after Holm correction; the closest, common_lag close-to-close, has an adjusted p-value of 0.07.

Market beta explains the raw signal, and the specific shock continues rather than reverts

This breakdown is exploratory: it was run after the pre-registered results. It adds the industry book's constraints one at a time, close-to-close.

Signal	Dollar only	+ beta	+ own lag	All three
conditional_signal	1.93% (3.0)	-0.28% (-0.6)	0.10% (0.2)	-0.70% (-1.5)
common_lag	6.63% (7.3)	2.32% (3.5)	2.83% (4.7)	1.19% (2.4)
shock_lag	1.24% (2.2)	1.37% (2.7)	1.11% (2.1)	1.32% (2.7)
leader_ret_lag	4.14% (5.8)	2.06% (3.5)	1.88% (3.5)	1.49% (3.1)

results/p4_exploratory_constraints.csv · return per year (Newey-West t), close to close

Beta neutrality alone takes the conditional signal from 1.9% a year ($t = 3.0$) to -0.3% ($t = -0.6$). The common component contains yesterday's market move, so ranking industries on it is largely a bet that high-beta industries keep moving with the market. The same constraint takes common_lag from 6.6% to 2.3%. The own-lag constraint removes part of what remains. Across industries, common_lag has a correlation of 0.63 with the followers' own previous-day return, because Part 2's common component is built from an index of those same followers. Part of what common_lag earns is therefore one-day industry momentum rather than information from the leader.

The shock leg behaves in the opposite way from Hypothesis 2. Its book earns between 1.1% and 1.4% a year under every set of constraints, so followers continue on the leader-specific shock instead of reverting. Because conditional_signal is common_lag minus shock_lag, it subtracts one continuing

predictor from another, and once beta is removed nothing is left. This is the portfolio-level counterpart of Part 2's wrong-signed coefficient on the shock.

The one survivor is too small to trade

The plain, unconditional leader return (`leader_ret_lag`) is the only signal that keeps a significant return after all three constraints: 1.49% a year close-to-close ($t = 3.1$) and 0.79% open-to-close ($t = 1.8$). Controlling for the reversal book leaves an open-to-close alpha of 1.0% a year ($t = 2.1$). That is about 0.3 bp a day, on a book that must buy and sell its whole gross every day to earn the open-to-close return.

Book	Breakeven, open to close (bp)	Breakeven, close to close (bp)	Median half-spread (bp)
<code>conditional_signal</code>	-0.09	-0.22	2.58
<code>common_lag</code>	0.14	0.23	2.58
<code>leader_ret_lag</code>	0.12	0.35	2.58
<code>shock_lag</code>	0.15	0.38	2.58
reversal (benchmark)	0.44	1.56	2.58

results/p4_costs.csv · 2007–2024 · one-way cost at which the book's gross return is zero

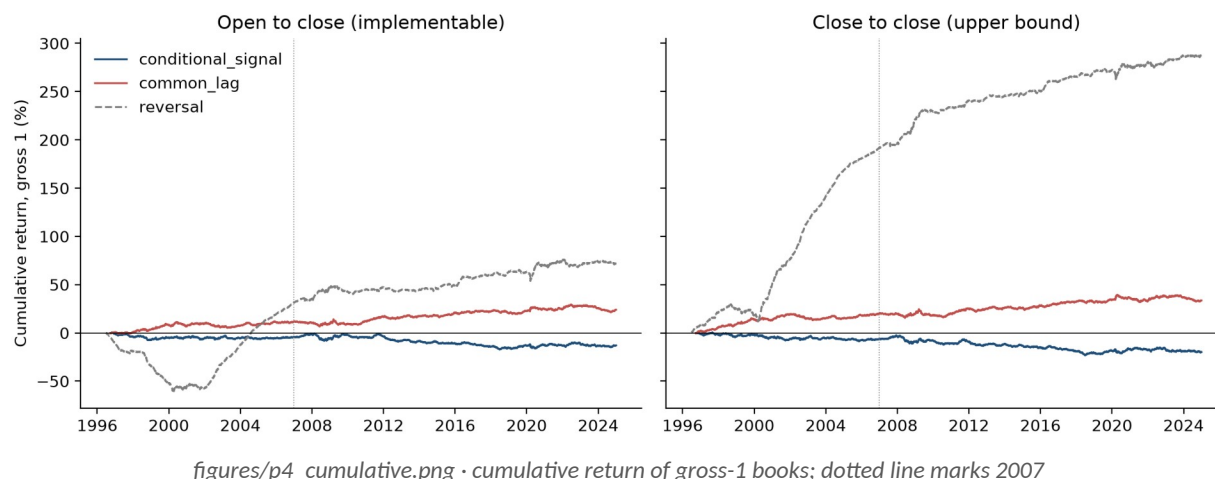
Every lead-lag book breaks even at under 0.4 bp, against a median half-spread of 2.6 bp for the stocks it holds, so trading costs exceed the return by roughly an order of magnitude.

Time patterns

Book	1996–2024	1996–2006	2007–2024	2010–2024	Last 18 m	Last 12 m
<code>conditional_signal</code>	-0.5% (-1.1)	-0.4% (-0.7)	-0.5% (-0.9)	-0.7% (-1.3)	0.3% (0.2)	0.5% (0.3)
<code>common_lag</code>	0.8% (1.9)	1.1% (1.6)	0.7% (1.2)	1.0% (1.6)	-2.2% (-1.3)	-1.3% (-0.6)
<code>leader_ret_lag</code>	0.8% (1.8)	1.1% (1.6)	0.6% (1.1)	0.9% (1.5)	-1.8% (-1.2)	-1.5% (-0.8)
reversal (benchmark)	2.5% (3.7)	3.0% (2.5)	2.2% (2.8)	1.9% (2.3)	-0.2% (-0.1)	-2.1% (-0.9)

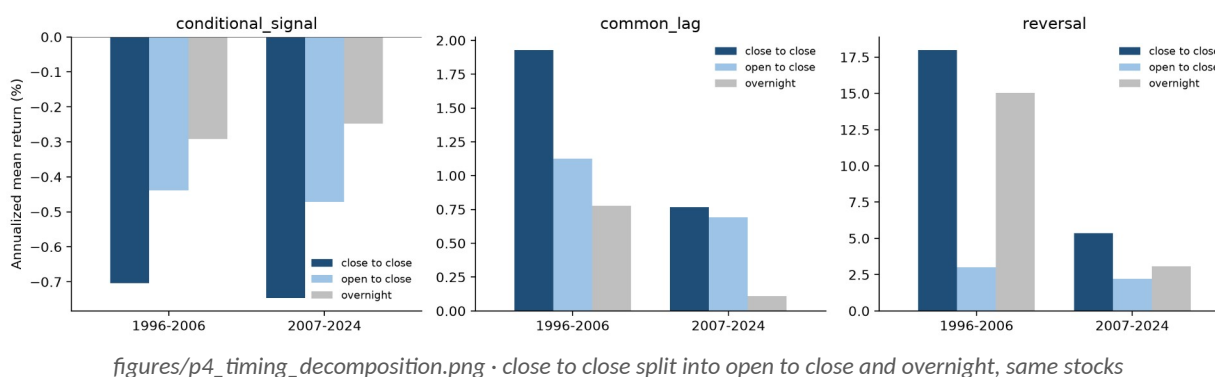
results/p4_performance.csv · open to close, return per year (Newey-West t); the data end in December 2024

`common_lag` and `leader_ret_lag` are stronger in 1996–2006 than afterwards, the same concentration Part 1 found, and both lose money over the last 18 months of the sample (July 2023 to December 2024). The recent windows are short: an 18-month Sharpe ratio has a standard error of about 0.8, so they are descriptive only.



The reversal benchmark is mostly bid-ask bounce and overnight catch-up

Part 3 found naive reversal to have by far the strongest rank IC. As a portfolio it is much weaker than it looks. Close-to-close it earns 10.1% a year (Sharpe 2.7), but on bid-ask midpoints it earns 1.9%, and in 1996–2006 it loses money on midpoints (–1.0% a year). Most of the close-to-close return is the bounce between bid and ask, and most of it arrives overnight: in 1996–2006, 15 of its 18 percentage points a year were earned between the close and the next open, which a trader cannot capture. Traded open-to-close it earns 2.5% a year, but it lost money in every year from 1996 to 2001, a 46% drawdown, before turning positive after decimalization in 2001. After 2007 it breaks even at 1.6 bp close-to-close and 0.4 bp open-to-close, below the half-spread in both cases.



Risk profile

The books carry almost no factor exposure. In regressions on the Fama-French five factors, momentum and the lagged market, R^2 is below 2% for every lead-lag book; for open-to-close books the market factor is an open-to-close market built from CRSP. common_lag, shock_lag and leader_ret_lag load negatively on the reversal book (t between –2.4 and –4.9), so part of their return is the opposite of short-term reversal. At gross 1 their volatility is low, 2.3% to 2.7% a year, which is why the headline's volatility overlay reaches its 3× cap and runs at 6.7% rather than the 10% target.

One-day 99% VaR from a trailing 250-day history averages 0.33% to 0.46% of gross, but it is breached too often: 96 to 115 times against about 69 expected, and the Kupiec test rejects for every book ($p <$

0.003). A trailing-window VaR adapts too slowly to the volatility regimes in these books. Stress periods are mild for the lead-lag books; the headline's worst is March 2020 (−2.4%).

Verdict: do not implement

Criterion (conditional_signal, open to close)	Result	Outcome
1. Mean return positive and significant after Holm correction	−0.46% a year; adjusted p = 0.28	Failed
2. Positive alpha after controlling for factors and the reversal book	−0.46% a year; p = 0.29	Failed
3. Breakeven cost above the median half-spread, 2007–2024	−0.09 bp against 2.58 bp	Failed

results/p4_decision.csv

The conditional push strategy fails all three pre-registered criteria, so the verdict is do not implement. This agrees with Part 3's corrected results, which also reject Hypothesis 3. At a one-day lag the signal does predict returns in a naive sort, but that predictability is a market-beta bet, the specific shock continues instead of reverting, and the one component that survives risk control, the unconditional leader move, is an order of magnitude too small to cover trading costs.

Using ChatGPT: a strong checklist, weaker on timing

Part 4 used ChatGPT twice, after the design was pre-registered, as an independent review; the prompts, outputs and evaluation are in `appendix/chatgpt_p4.md`. Asked how this backtest could show a profit that is not real, it listed about 70 failure modes. Without being told, it worked out the issues that decide this section: a signal shared by every follower in an industry turns the strategy into industry allocation, ranking on a component that contains yesterday's market move is a beta bet, and the industry index used to decompose the leader contains the very followers being predicted. It also said to lag on the trading calendar rather than the previous observation, which is the defect the code audit found in the team's code.

Its weakest point was timing. Our prompt said we trade at the close; it correctly ruled out trading at the same close that produced the signal and moved the trade to the next close, a full session late. That is the two-session gap of the first version of Part 3, and under it every lead-lag book here earns roughly nothing (`leader_ret_lag t = 0.02`). It suggested testing entry at the next open but did not make it the headline, which is partly a lesson about our prompt: it should have asked for the earliest feasible execution rather than stating one. On construction it imposed the same three constraints as Part 4, but at the stock level, so an industry's weight grows with its number of followers, a problem it had itself flagged in its first answer. Its recommendation of a factor risk model points to a real gap: Part 4 has none. As in HW1 and HW2, it was good at listing what could go wrong and weak at judging which items matter for this dataset.

Limitations

Costs are assessed as a breakeven against quoted half-spreads; Part 5 applies a full cost model. The industry books rest on about 48 independent bets a day, so a stock-level risk model would understate

their risk; they are sized by gross exposure and realized volatility instead of a full factor risk model. The Fama-French factors exist only close-to-close, so the open-to-close attribution is approximate beyond the market factor. No out-of-sample period was reserved in 1996–2024; that test belongs to Part 5.